

Chapter 2: The first law of thermodynamics

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Summary of Chapter I

- ✓ In thermodynamics, the universe is divided into the **system** (that part of the universe of interest to us) and the **surroundings**. There are several kinds of walls between the system and the surroundings, and each type gives rise to a system with specific characteristics.
- ✓ In thermodynamics, the **equilibrium** of the system is of interest in that, if it is known, one can determine if the state of the system will change and in which direction such a change would go.

Summary of Chapter I

- ✓ The **state** of a simple system is determined by its **temperature** and pressure, the two intensive **independent variables** of the system.
- ✓ Other thermodynamic variables are functions of pressure and temperature, and graphs may be plotted which display the equilibrium states of the system as a function of the independent variables.

Summary of Chapter I

- ✓ The Zeroth Law of Thermodynamics introduces the intensive variable **temperature, T** .
- ✓ The First Law of Thermodynamics states that the energy of the universe is constant, shows that different forms of energy may be converted into one another, and introduces the extensive thermodynamic variable **internal energy, U** .
- ✓ The Second Law of Thermodynamics defines which processes may occur spontaneously and introduces the extensive thermodynamic variable **entropy, S** . **The entropy of the universe can never decrease.**
- ✓ The Third Law of Thermodynamics states that all aspects of entropy approach the value of zero as the temperature of the system approaches zero, if the system is in complete internal equilibrium.

Concepts and terms introduced in Chapter 1

Alloy

Boundaries: adiabatic/diathermal/permeable/semipermeable

Boyle's law

Charles' law

Coefficient of thermal expansion

Components (Concentration, Composition)

Conservation of energy

Conversion of energy

Energy/work

Equilibrium phase diagram

Exact differentials

Extensive/intensive thermodynamic variables

Gas constant, R

Homogeneous/heterogeneous systems

Concepts and terms introduced in Chapter 1

Ideal gases/ideal gas law

Independent/dependent thermodynamic variables

Isolated/closed/open systems

Isothermal compressibility

Laws of Thermodynamics

Microscopic/macroscopic thermodynamic variables

Solid/liquid solutions

System/surroundings

Thermodynamic state

Thermodynamic state functions

Triple point

Vapor pressure

V - P - T space

Concepts and terms introduced in Chapter 1

- ✓ In chemistry, **concentration** is the abundance of a constituent divided by the total volume of a mixture. Several types of mathematical description can be distinguished: mass concentration, molar concentration, number concentration, and volume concentration. A concentration can be any kind of chemical mixture, but most frequently solutes and solvents in solutions.
- ✓ In phase-diagrams, **components** are chemical species of fixed composition. Chemical **composition** refers to the identity and relative number of the chemical elements that make up any particular compound.



Concepts and terms introduced in Chapter 1

Quantitative notation [\[edit \]](#)

There are four quantities that describe concentration:

Mass concentration [\[edit \]](#)

Main article: [Mass concentration \(chemistry\)](#)

The mass concentration ρ_i is defined as the [mass](#) of a constituent m_i divided by the volume of the mixture V :

$$\rho_i = \frac{m_i}{V}.$$

The [SI unit](#) is kg/m³ (equal to g/L).

Molar concentration [\[edit \]](#)

Main article: [Molar concentration](#)

The molar concentration c_i is defined as the [amount](#) of a constituent n_i (in moles) divided by the volume of the mixture V :

$$c_i = \frac{n_i}{V}.$$

The [SI unit](#) is mol/m³. However, more commonly the unit mol/L (= mol/dm³) is used.

Number concentration [\[edit \]](#)

Main article: [Number concentration](#)

The number concentration C_i is defined as the number of entities of a constituent N_i in a mixture divided by the volume of the mixture V :

$$C_i = \frac{N_i}{V}.$$

The [SI unit](#) is 1/m³.

Volume concentration [\[edit \]](#)

The **volume concentration** σ_i (not to be confused with [volume fraction](#)^[2]) is defined as the volume of a constituent V_i divided by the volume of the mixture V :

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2.1. Introduction

- ✓ The First Law of Thermodynamics is sometimes thought to be merely an extension to **the Law of Conservation of Energy**.
- The First Law introduces the important thermodynamic state variable **internal energy**, U , and the law posits that energy may be converted from one of its forms to another form.
- The First Law introduces the important concept that the transfer of thermal energy (**heat**) is a different kind of energy than that which is done during a process of **work**.

2.1. Introduction

- ✓ **Kinetic energy** is conserved in a frictionless system of interacting rigid elastic bodies. A collision between two of these bodies results in a transfer of kinetic energy from one to the other, the work done by the one equals the work done on the other, and the total kinetic energy of the system is unchanged as a result of the collision.
- ✓ If, friction occurs in the system, then with continuing collision and interaction among the bodies, the total dynamic energy of the system decreases and heat is produced. It is reasonable to expect that a relationship exists between the dynamic energy dissipated and the heat produced as a result of the effects of friction.

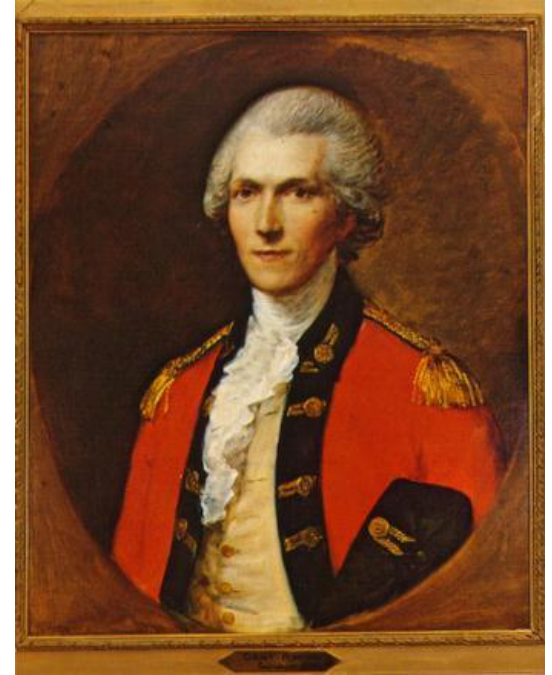


The internal energy U and the enthalpy H

Thermodynamics of Materials & Phase transformations

2.2. The relationship between heat and work

- ✓ Count Rumford (1798): during the boring of cannon, he noticed that the heat produced during the boring was roughly proportional to the work performed during the boring. In this sense, heat had been regarded as being an invisible fluid called *caloric* which resided between the constituent particles of a substance.



- ✓ In the caloric theory of heat, the temperature of a substance was considered to be determined by the quantity of *caloric gas* which it contained, and two bodies of differing temperature, when placed in contact with one another, came to an intermediate common temperature as the result of caloric flowing between them.

2.2. The relationship between heat and work

- ✓ In 1799, Humphrey Davy melted two blocks of ice by rubbing them together in a vacuum. In this experiment the latent heat necessary to melt the ice was provided by the mechanical work performed in rubbing the blocks together.



Humphrey Davy

- ✓ From 1840, James Joule carried out a series of experiments, e.g., work was performed in a certain quantity of adiabatically contained water and measured the resultant increase in the temperature of the water. The relationship between heat and work was then placed on a firm quantitative basis.



James Joule

2.2. The relationship between heat and work

James Joule observed that a direct proportionality existed between the work done and the resultant increase in temperature and that the same proportionality existed no matter what means were employed in the work production. Methods of work production used by Joule included.

- Rotating a paddle wheel immersed in the water
- An electric motor driving a current through a coil immersed in the water
- Compressing a cylinder of gas immersed in the water
- Rubbing together two metal blocks immersed in the water

Work  **Heat**

2.2. The relationship between heat and work

- ✓ This proportionality gave rise to the notion of a *mechanical equivalent of heat*, and for the purpose of defining this figure it was necessary to define a unit of heat.
- ✓ This unit is the calorie (or 15 °calorie), which is the quantity of heat required to increase the temperature of 1 gram of water from 14.5 °C to 15.5 °C. On the basis of this definition Joule determined the value of the mechanical equivalent of heat to be 0.241 calories per joule.
- ✓ The presently accepted value is 0.2389 calories (15 °C calories) per joule.

2.3 Internal energy and the first law of thermodynamics

- ✓ Joule's experiments resulted in the statement that “the change of a body inside an adiabatic enclosure from a given initial state to a given final state involves the same amount of work by whatever means the process is carried out.”
- ✓ The statement is a preliminary formulation of the First Law of Thermodynamics, and in view of this statement, it is necessary to define some function which depends only on the internal state of a body or system. Such a function is U , the internal energy.

2.3 Internal energy and the first law of thermodynamics

- ◆ When a body of mass m is lifted in a gravitational field from height h_1 to height h_2 , the work w done on the body is given by:

$$w = \text{force} \times \text{distance}$$

$$= mg \times (h_2 - h_1)$$

$$= mgh_2 - mgh_1$$

$$= \text{potential energy at position } h_2$$

$$\text{minus potential energy at position } h_1$$

- As the potential energy of the body of given mass m depends only on the position of the body in the gravitational field, it is seen that the work done on the body is dependent only on its final and initial positions and is independent of the path taken by the body between the two positions, i.e., between the two states.

2.3 Internal energy and the first law of thermodynamics

- ◆ The application of a force f to a body of mass m causes the body to accelerate according to Newton's Law:

$$f = ma = m \frac{dv}{dt}$$

The work done on the body is thus obtained by integrating

$$dw = fdl = m \frac{dv}{dt} dl = mv dv$$

$$w = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$$

= the kinetic energy of the body at velocity v_2 (state 2)

minus the kinetic energy of the body at velocity v_1 (state 1)

2.3 Internal energy and the first law of thermodynamics

- ✓ The work done on the body is the difference between the values of a function of the state of the body and is independent of the path taken by the body between the states.
- ✓ In the case of work being done on an adiabatically contained body of constant potential and kinetic energy, the pertinent function which describes the state of the body, or the change in the state of the body, is the internal energy U .
- ✓ The work done on, or by, an adiabatically contained body equals the change in the internal energy of the body, i.e., equals the difference between the value of U in the final state and the value of U in the initial state.

2.3 Internal energy and the first law of thermodynamics

- ✓ In describing work, it is conventional to assign a negative value to work done on a body and a positive value to work done by a body.
- ✓ This convention arises because, when a gas expands, and hence does work against an external pressure, the integral $\int_1^2 P dV$, which is the work performed, is a positive quantity.
- ✓ For an adiabatic process in which work w is done on a body, as a result of which its state moves from A to B.

$$w = -(U_B - U_A)$$

- If work w is done on the body, then $U_B > U_A$ and if the body itself performs work, then $U_B < U_A$.

2.3 Internal energy and the first law of thermodynamics

- ✓ In Joule's experiments the change in the state of the adiabatically contained water was measured as an increase in the temperatures of the water.
- ✓ The same increase in temperature, and hence the same change of state, could have been produced by placing the water in thermal contact with a source of heat and allowing heat q to flow into the water.
- ✓ In describing heat changes it is conventional to assign a negative value to heat which flows out of a body (an **exothermic** process) and a positive value to heat which flows into a body (an **endothermic** process).

$$q = U_B - U_A$$

- Thus, when heat flows into the body, q is a positive quantity and $U_B > U_A$, whereas if heat flows out of the body, $U_B < U_A$ and q is a negative quantity.

2.3 Internal energy and the first law of thermodynamics

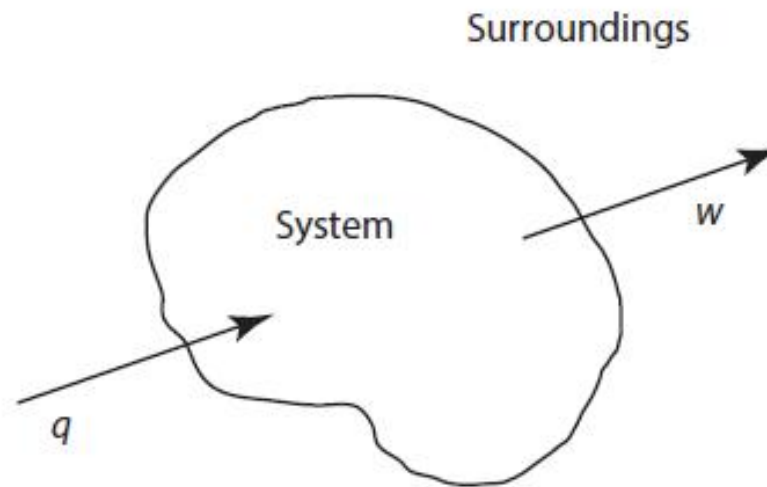


Figure 2.1 Schematic showing the sign convention for the work, w (positive), done by the system and energy transferred as heat, q , added to the system.

$$q = U_B - U_A$$

$$q > 0, U_B > U_A$$

$$q < 0, U_B < U_A$$

$$w = -(U_B - U_A)$$

$$w > 0, U_B < U_A$$

$$w < 0, U_B > U_A$$

2.3 Internal energy and the first law of thermodynamics

- ✓ Consider a body, initially in the state A, which performs work w , absorbs heat q , and, as a consequence, moves to the state B.
- ✓ The absorption of heat q *increases* the internal energy of the body by the amount q , and the performance of work w by the body *decreases* its internal energy by the amount w . Thus the total change in the internal energy of the body is:

$$\Delta U = U_B - U_A = q - w$$

$$dU = \delta q - \delta w$$

The first law of thermodynamics

2.3 Internal energy and the first law of thermodynamics

$$dU = \delta q - \delta w$$

- ✓ As U is a state function, the integration of dU between two states gives a value which is independent of the path taken by the system between the two states.
- ✓ The heat and work effects, which involve energy in transit, depend on the path taken between the two states, as a result of which the integrals of δw and δq cannot be evaluated without a knowledge of the path.
- The use of the symbol “ d ” indicates a differential element of a state function or state property, the integral of which is independent of the path, and use of the symbol “ δ ” indicates a differential element of some quantity which is not a state function.

2.3 Internal energy and the first law of thermodynamics

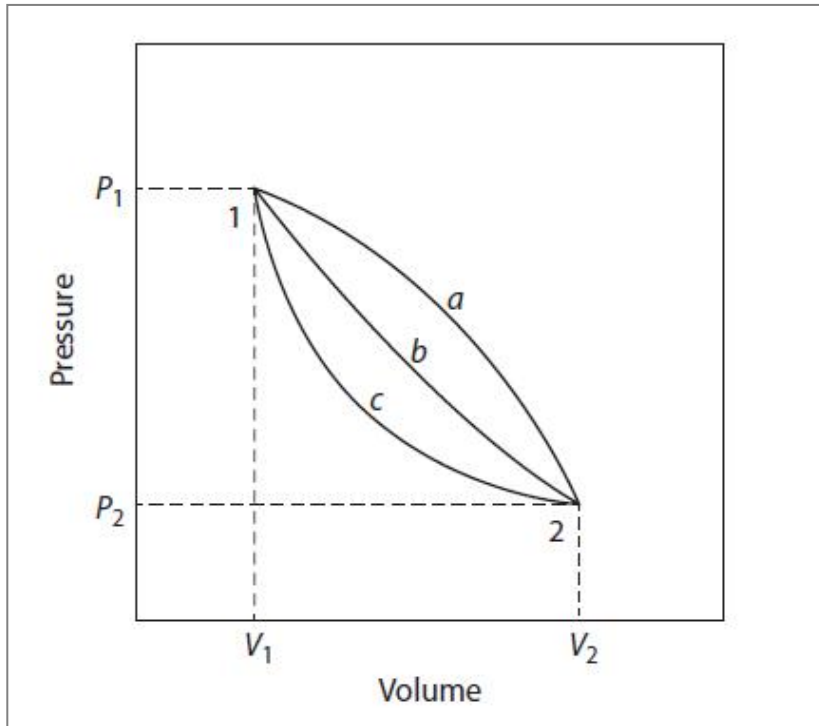


Figure 2.2 Three process paths taken by a fixed quantity of gas in moving from the state 1 to the state 2, showing that the work performed by the gas during expansion depends on the path traversed during the process.

$$\Delta U = \int_1^2 dU + \int_2^1 dU = (U_2 - U_1) + (U_1 - U_2) = 0$$



- ✓ In the case of a cyclic process which returns the system to its initial state, e.g., the process $1 \rightarrow 2 \rightarrow 1$, the change in U as a result of this process is zero.
- ✓ The vanishing of a cyclic integral is a property of a state function

2.3 Internal energy and the first law of thermodynamics

- ✓ As U is a state function, then for a simple system consisting of a given amount of substance of fixed composition, the value of U is fixed once any two properties (the independent variables) are fixed. If temperature and volume are chosen as the independent variables, then

$$U = U(V, T) \quad dU = \left(\frac{\partial U}{\partial V} \right)_T dV + \left(\frac{\partial U}{\partial T} \right)_V dT$$

- ✓ It is of interest to examine those processes which can occur when the value of one of the independent variables is maintained constant and the other is allowed to vary:
 - V is maintained constant, **isochore** or **isometric** processes
 - P is maintained constant, **isobaric** processes
 - T is maintained constant, **isothermal** processes
 - $q=0$, **adiabatic** processes

2.4 Constant-volume processes

- ✓ If the volume of a system is maintained constant during a process, then the system does no work ($\int P dV=0$), and from the First Law:

$$dU = \delta q - \delta w \quad \boxed{w=0} \Rightarrow dU = \delta q_V$$

$$\Delta U = U_B - U_A = q - w \quad \boxed{w=0} \Rightarrow \Delta U = q_V$$

For such a process, which shows that the increase or decrease in the internal energy of the system equals, respectively, the heat absorbed or rejected by the system during the process.

Isochoric process

2.5 Constant-pressure processes and the enthalpy H

- ✓ If the pressure is maintained constant during a process which takes the system from state 1 to state 2, then the work done by the system is given as:

$$w = \int_1^2 P dV = P \int_1^2 dV = P(V_2 - V_1)$$

The First Law gives:

$$U_2 - U_1 = q_P - w = q_P - P(V_2 - V_1)$$

$$(U_2 + PV_2) - (U_1 + PV_1) = q_P$$

Isobaric process

2.5 Constant-pressure processes and the enthalpy H

$$(U_2 + PV_2) - (U_1 + PV_1) = q_P$$

- ✓ As the expression $(U+PV)$ contains only state functions, the expression itself is a state function. This is termed the **enthalpy**, H , i.e.,:

$$H = U + PV$$

$$(U_2 + P_2V_2) - (U_1 + P_1V_1) = H_2 - H_1 = q_P$$

The enthalpy change during a constant-pressure process is equal to the heat admitted to or withdrawn from the system during the process.

2.6 Heat capacity

- ✓ The heat capacity, C , of a system is the ratio of the heat added to or withdrawn from the system to the resultant change in the temperature of the system.

$$C = \frac{q}{\Delta T}$$

$$C = \frac{\delta q}{\delta T}$$

- The concept of heat capacity is only used when the addition of heat to or withdrawal of heat from the system produces a temperature change; the concept is not used when a phase change is involved.
- For example, if the system is a mixture of ice and water at 1 atm pressure and 0 °C, then the addition of heat simply melts some of the ice and no change in temperature occurs. In such a case the heat capacity, as defined, would be infinite.

2.6 Heat capacity

$$C_V = \left(\frac{\delta q}{dT} \right)_V \quad \text{The heat capacity at constant volume}$$

$$C_P = \left(\frac{\delta q}{dT} \right)_P \quad \text{The heat capacity at constant pressure}$$

$$C_V = \left(\frac{\delta q}{dT} \right)_V = \left(\frac{dU}{dT} \right)_V \quad dU = C_V dT$$

$$C_P = \left(\frac{\delta q}{dT} \right)_P = \left(\frac{dH}{dT} \right)_P \quad dH = C_P dT$$

$$dU = \delta q - \delta w$$

2.6 Heat capacity

- ✓ The heat capacity, being dependent on the size of the system, is an extensive property.
- ✓ In normal usage it is more convenient to use the heat capacity per unit quantity of the system.
- ✓ The **specific heat** of the system is the heat capacity per gram at constant P , and the **molar heat capacity** is the heat capacity per mole at constant pressure or at constant volume.

For a system containing n moles:

$$C_V = nc_V$$

$$C_P = nc_P$$

2.6 Heat capacity

- ✓ It is to be expected that, for any substance, C_p will be of greater magnitude than C_v .
- ✓ If it is required that the temperature of a system is increased by a certain amount, then, if the process is carried out at a constant volume, all of the heat added is used solely to raise the temperature of the system.
- ✓ If the process is carried out at constant pressure, then, in addition to raising the temperature by the required amount, the heat added is required to provide the work necessary to expand the system at the constant pressure. This work of expansion against the constant pressure per degree of temperature increase is calculated as

$$\frac{PdV}{dT} \quad \text{or} \quad P\left(\frac{\partial V}{\partial T}\right)_P$$

2.6 Heat capacity

$$\frac{PdV}{dT} \text{ or } P\left(\frac{\partial V}{\partial T}\right)_P \quad \rightarrow \quad c_P - c_V = P\left(\frac{\partial V}{\partial T}\right)_P$$

$$H = U + PV$$

$$c_P = \left(\frac{\delta q}{dT}\right)_P = \left(\frac{dH}{dT}\right)_P = \left(\frac{\partial U}{\partial T}\right)_P + P\left(\frac{\partial V}{\partial T}\right)_P$$

$$c_V = \left(\frac{\delta q}{dT}\right)_V = \left(\frac{dU}{dT}\right)_V = \left(\frac{\partial U}{\partial T}\right)_V$$

$$c_P - c_V = \left(\frac{\partial U}{\partial T}\right)_P + P\left(\frac{\partial V}{\partial T}\right)_P - \left(\frac{\partial U}{\partial T}\right)_V$$

2.6 Heat capacity

$$c_P - c_V = \left(\frac{\partial U}{\partial T} \right)_P + P \left(\frac{\partial V}{\partial T} \right)_P - \left(\frac{\partial U}{\partial T} \right)_V \quad \longleftrightarrow \quad c_P - c_V = P \left(\frac{\partial V}{\partial T} \right)_P$$

$$U = U(V, T) \quad \Rightarrow \quad dU = \left(\frac{\partial U}{\partial V} \right)_T dV + \left(\frac{\partial U}{\partial T} \right)_V dT$$

$$\left(\frac{\partial U}{\partial T} \right)_P = \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P + \left(\frac{\partial U}{\partial T} \right)_V$$

$$\begin{aligned} c_P - c_V &= \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P + \left(\frac{\partial U}{\partial T} \right)_V + P \left(\frac{\partial V}{\partial T} \right)_P - \left(\frac{\partial U}{\partial T} \right)_V \\ &= \left(\frac{\partial V}{\partial T} \right)_P \left[P + \left(\frac{\partial U}{\partial V} \right)_T \right] \end{aligned}$$

2.6 Heat capacity

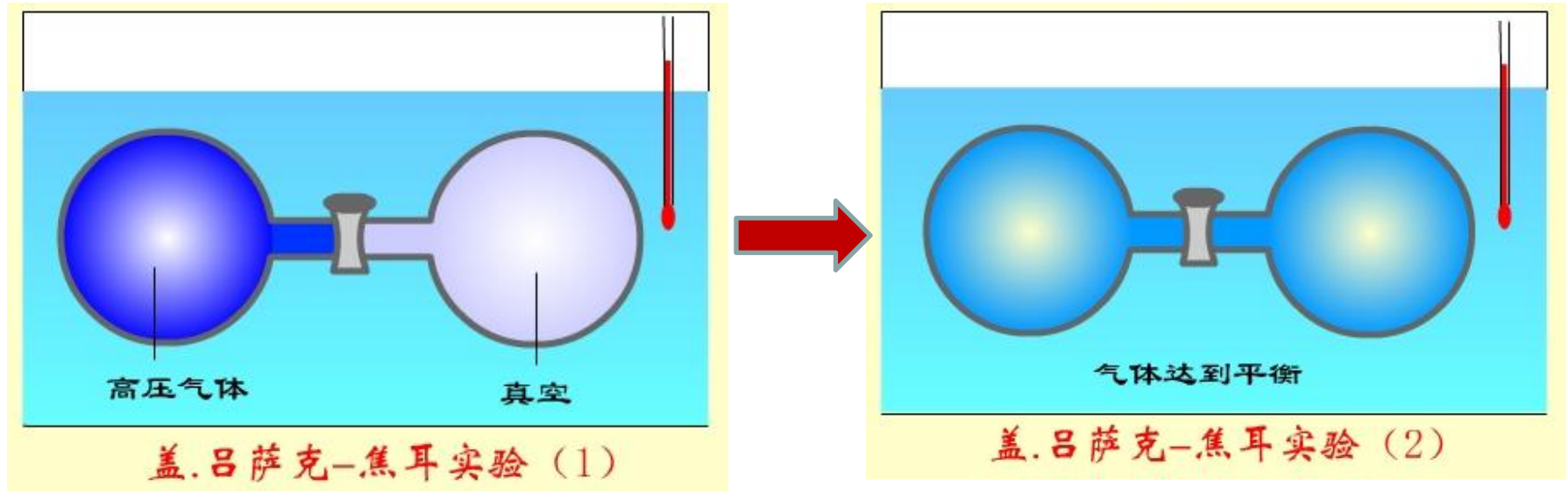
$$c_P - c_V = P \left(\frac{\partial V}{\partial T} \right)_P$$



$$c_P - c_V = \left(\frac{\partial V}{\partial T} \right)_P \left[P + \left(\frac{\partial U}{\partial V} \right)_T \right]$$

- ✓ Joule performed an experiment which involved filling a copper vessel with a gas at some pressure and connecting this vessel via a stopcock to a similar but evacuated vessel.
- ✓ The two-vessel system was immersed in a quantity of adiabatically contained water and the stopcock was opened, thus allowing free expansion of the gas into the evacuated vessel.

2.6 Heat capacity



Neither the gas nor the bath temperature is changed: $Q = 0$

The system does not do work: $W = 0$

According to the first law: $\Delta U = 0$

2.6 Heat capacity

$$dU = \left(\frac{\partial U}{\partial V} \right)_T dV + \left(\frac{\partial U}{\partial T} \right)_V dT = 0$$

- ✓ As $dT=0$ (experimentally determined) and $dV \neq 0$ then the term $(\partial U / \partial V)_T$ must be zero. Joule thus concluded that the internal energy of a gas is a function only of temperature and is independent of the volume (and hence pressure).

- For a gas:
$$c_P - c_V = P \left(\frac{\partial V}{\partial T} \right)_P$$
- For an ideal gas $PV=RT$:
$$c_P - c_V = R$$

$$\left(\frac{\partial U}{\partial V} \right)_T = 0$$

$$\left(\frac{\partial U}{\partial P} \right)_T = 0$$

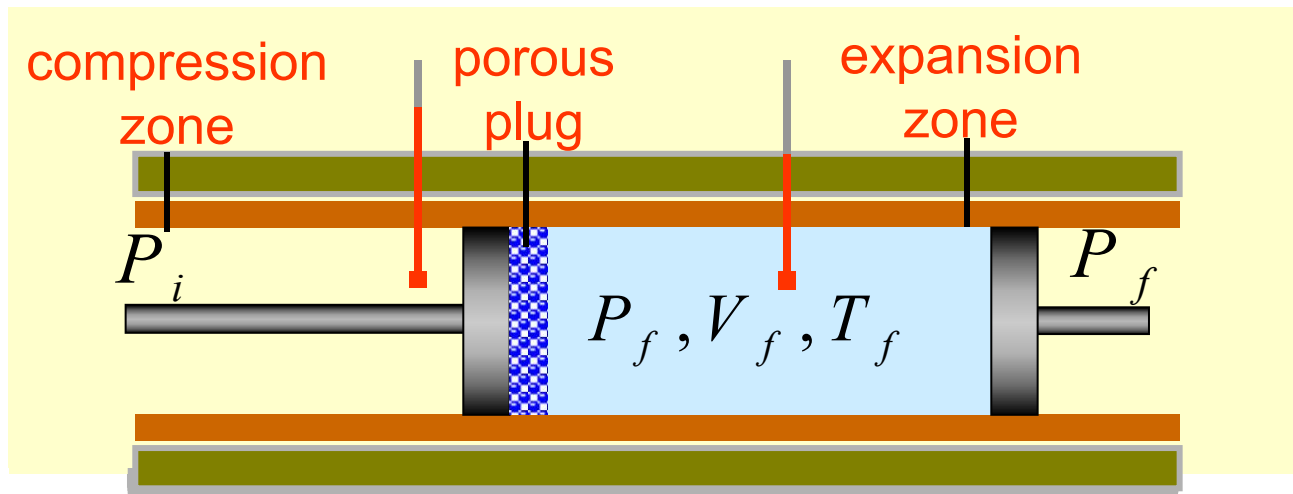
$$\left(\frac{\partial H}{\partial P} \right)_T = 0$$

$$\left(\frac{\partial H}{\partial V} \right)_T = 0$$

2.6 Heat capacity

- ✓ Joule's experiment on free expansion of gas in 1843 was not precise enough. In 1852 Joule and Thomson designed a new experiment called the throttle process.
- ✓ With this experiment, people understood more about the properties of U and H of real gases. It has important applications in obtaining cryogenic state and gas liquefaction.

Throttle process



2.6 Heat capacity

- ✓ The throttle process is carried out in a adiabatic cylinder, $Q = 0$, so:

$$U_2 - U_1 = \Delta U = W$$

- ✓ At the beginning, the work done by the surroundings due to compression of a certain amount of gas (that is, the work obtained when gas is the system) is:

$$W_1 = -P_1 \Delta V = -P_1 (0 - V_1) = P_1 V_1$$

- ✓ Gas expands through the hole, the work done on surroundings is:

$$W_2 = -P_2 \Delta V = -P_2 (V_2 - 0) = -P_2 V_2$$

2.6 Heat capacity

- ✓ When compression and expansion, the change of the net work in the system is the algebraic sum of the two kinds of work.

$$W = W_1 + W_2 = P_1V_1 - P_2V_2$$

$$U_2 - U_1 = \Delta U = W = P_1V_1 - P_2V_2$$

$$U_2 + p_2V_2 = U_1 + p_1V_1$$

$$H_2 = H_1 \quad \Delta H = 0$$

The throttle process is an **isenthalpic** process.

2.6 Heat capacity

- ✓ In a more critical experiment performed by Joule and Thomson, in which an adiabatically contained gas of molar volume V_1 at the pressure P_1 was throttled through a porous diaphragm to the pressure P_2 and the molar volume V_2 , a change in the temperature of the gas was observed, which showed that, for real gases, $(\partial U/\partial V)_T \neq 0$.
- ✓ The reason for Joule's not observing a temperature rise in the original experiment was that the *heat capacity of the copper vessels and the water was considerably greater than the heat capacity of the gas*, and thus the small heat changes which actually occurred in the gas were absorbed in the copper vessels and the water. This decreased the actual temperature change to below the limits of the then-available means of temperature measurement.

2.6 Heat capacity

$$c_P - c_V = \left(\frac{\partial V}{\partial T} \right)_P \left[P + \left(\frac{\partial U}{\partial V} \right)_T \right]$$

$$P \left(\frac{\partial V}{\partial T} \right)_P$$

The work done by the system per degree rise in temperature in expanding against the constant external pressure P acting on the system.

$$\left(\frac{\partial V}{\partial T} \right)_P \left(\frac{\partial U}{\partial V} \right)_T$$

The work done per degree rise in temperature in expanding against the internal cohesive forces acting between the constituent particles of the substance.

2.6 Heat capacity

- ✓ An ideal gas is a gas consisting of non-interacting particles, and hence, the atoms or molecules of an ideal gas can be separated from one another without the expenditure of work. Thus, for an ideal gas.

$$\left(\frac{\partial V}{\partial T}\right)_P = \alpha V \neq 0 \quad \Rightarrow \quad \left(\frac{\partial V}{\partial T}\right)_P \left(\frac{\partial U}{\partial V}\right)_T = 0$$

$$\Rightarrow \quad \left(\frac{\partial U}{\partial V}\right)_T = 0$$

- In real gases the internal pressure contribution is very much smaller in magnitude than the external pressure contribution
- In liquids and solids, in which the interatomic forces are considerable, the work done in expanding the system against the external pressure is insignificant in comparison with the work done against the internal pressure. Thus for liquids and solids the term is very large.

2.6 Heat capacity

- ✓ ΔU in real gas is not only related to the temperature, but also is related to the volume (or pressure).

$$dU = \left(\frac{\partial U}{\partial T} \right)_V dT + \left(\frac{\partial U}{\partial V} \right)_T dV$$

- ✓ Because of the interaction between molecules of real gas, the change in thermodynamic energy can be measured by the energy dissipated against the intermolecular attraction during isothermal expansion.

$\left(\frac{\partial U}{\partial V} \right)_T$ is called the internal pressure, that is: $P_{Inter} = \left(\frac{\partial U}{\partial V} \right)_T$

For isothermal condition, $dU = P_{Inter} dV$

Thermodynamics of Materials & Phase transformations

The End



2.6 Heat capacity

$$dU = \left(\frac{\partial U}{\partial T} \right)_V dT + \left(\frac{\partial U}{\partial V} \right)_T dV$$



Divided by dT

$$\frac{dU}{dT} = \left(\frac{\partial U}{\partial T} \right)_V + \left(\frac{\partial U}{\partial V} \right)_T \frac{dV}{dT}$$



$$\left(\frac{\partial U}{\partial T} \right)_P = \left(\frac{\partial U}{\partial T} \right)_V + \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P$$

2.6 Heat capacity

Prove:
$$\left(\frac{\partial U}{\partial T}\right)_P = \left(\frac{\partial U}{\partial T}\right)_V + \left(\frac{\partial U}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_P$$

Suppose: $U = U(T, V), \quad V = V(T, P)$

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV$$

$$dV = \left(\frac{\partial V}{\partial T}\right)_P dT + \left(\frac{\partial V}{\partial P}\right)_T dP$$

Substituting the expression of dV , we have:

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T \left[\left(\frac{\partial V}{\partial T}\right)_P dT + \left(\frac{\partial V}{\partial P}\right)_T dP \right]$$

2.6 Heat capacity

Rearranging to separate dP from dT , we have:

$$\begin{aligned} dU &= \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial P} \right)_T dP + \left[\left(\frac{\partial U}{\partial T} \right)_V + \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P \right] dT \\ &= \left(\frac{\partial U}{\partial P} \right)_T dP + \left[\left(\frac{\partial U}{\partial T} \right)_V + \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P \right] dT \end{aligned}$$

Because U is a function of T and P . $U = U(T, P)$

$$dU = \left(\frac{\partial U}{\partial P} \right)_T dP + \left(\frac{\partial U}{\partial T} \right)_P dT$$

According to the two expression of dU , we have:

$$\left(\frac{\partial U}{\partial T} \right)_P = \left(\frac{\partial U}{\partial T} \right)_V + \left(\frac{\partial U}{\partial V} \right)_T \left(\frac{\partial V}{\partial T} \right)_P$$